GRADUATION PROJECT | DEPI – DIGITAL EGYPT PIONEERS INITIATIVE

Arkan Home

E-Commerce Data Localization & Business Intelligence Analytics

An end-to-end analytics project transforming raw e-commerce data into a localized Egyptian business intelligence solution using Excel, SQL Server, Star Schema modeling, DAX, and Power BI.

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EXCEL | DATA CLEANING | DATA VALIDATION | SQL SERVER | ETL | STAR SCHEMA | DAX | POWER BI | DASHBOARD DESIGN | BUSINESS INSIGHTS

PROJECT OVERVIEW

Project Executive Summary

From raw e-commerce data to localized business intelligence insights

Arkan Home is an end-to-end analytics project that transforms the public Olist e-commerce dataset into a localized Egyptian business case for a simulated Home Decor and Furniture company.

Business Context

Localized the original e-commerce dataset into an Egyptian market scenario focused on Home Decor and Furniture business performance.

Data Engineering

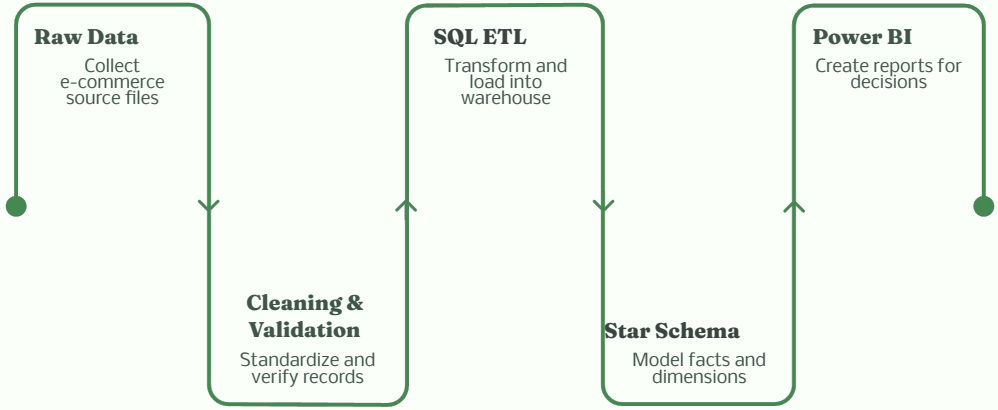
Prepared, cleaned, validated, and transformed raw transactional data using Excel, SQL Server, and structured ETL logic.

Analytical Modeling

Built a Star Schema model with Fact_Sales and dimension tables to support accurate, scalable, and business-ready analytics.

Business Insights

Delivered an interactive Power BI dashboard covering executive performance, sales, logistics, customer behavior, and growth analysis.



Key Message: We did not just build a dashboard. We built a complete business intelligence solution.

DATA FOUNDATION

Dataset & Localization Strategy

How we transformed the original Olist data into an Egyptian business case

The original Brazilian e-commerce dataset was transformed into a localized Egyptian business scenario to make the analysis more market-relevant and business-oriented.

Source Dataset

Brazilian Olist E-Commerce Dataset. Main source tables: Orders, Order Items, Payments, Reviews, Products, Customers, Sellers.

Localization Assets

- Geography_Mapping.xlsx
- product_category_mapping.csv
- State Flags.csv

Localization Logic

- Mapped original geography to Egyptian cities and regions
- Reclassified product categories into Home Decor & Furniture groups
- Adapted the business scenario to fit a realistic Egyptian e-commerce case

Key Message: We did not just analyze a dataset. We localized it into a market-specific business case.

DATA ENGINEERING

SQL Data Pipeline

From raw staging tables to a business-ready analytical model

The SQL layer was built as a controlled data pipeline, starting with raw staging tables, followed by cleaning, validation, ETL transformation, and star schema modeling.

1

Raw Table Setup

Created the initial raw layer inside SQL Server under **Home_Arkan_DB** to receive the original source entities in a structured staging environment.

2

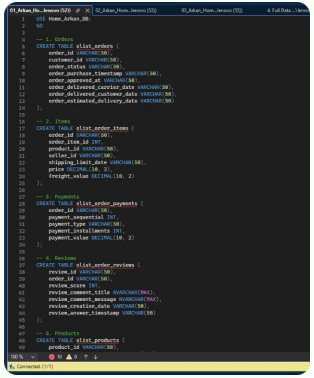
Cleaning & Validation

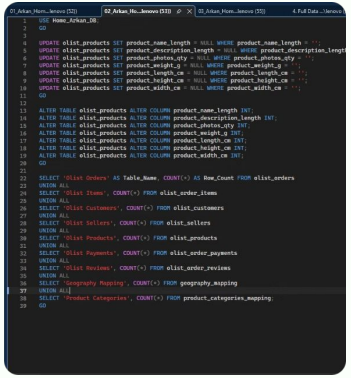
Replaced blank values, standardized data types, and verified row counts across core transactional and mapping tables.

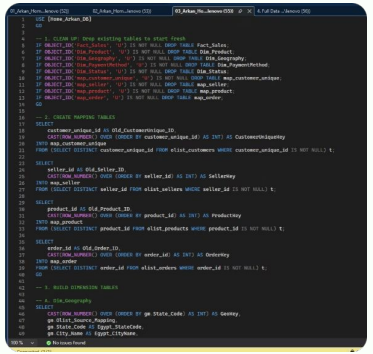
3

ETL & Star Schema

Built mapping tables, dimension tables, and the central **Fact_Sales** table to prepare the data for scalable Power BI reporting.







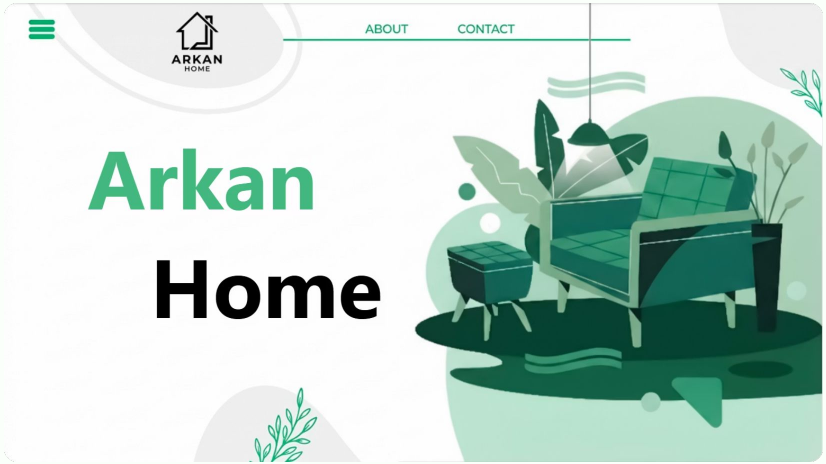
Key Message: The SQL pipeline transformed raw data into a trusted analytical foundation. This layer prepared the project for accurate modeling, DAX measures, and Power BI reporting.

POWER BI ARCHITECTURE

Power BI Dashboard Architecture

Connecting model design, navigation, and interactive business reporting

The Power BI solution was designed as a connected analytical experience, combining a branded landing page, structured data model, interactive navigation, slicers, and dashboard pages.



Dashboard Ecosystem

- Landing Page
- Executive Overview
- Sales
- Logistics
- Customer Behavior
- Growth Analysis

Model Foundation

- Central Fact_Sales table
- Connected dimension tables
- Consistent relationship logic
- Clean filtering and scalable reporting

Interactivity & UX

- Page navigation
- Year and month slicers
- Cross-filtering across visuals
- Help overlay and guided user flow

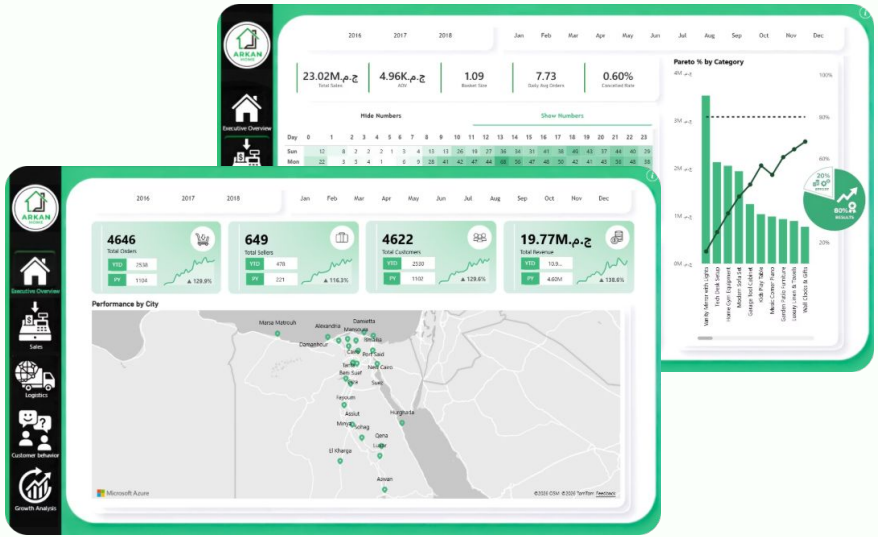
Key Message: The Power BI layer converted the analytical model into a guided business intelligence experience. It connected navigation, visuals, and business domains into one decision-support system.

BUSINESS INTELLIGENCE

Executive & Sales Insights

Combining leadership overview with commercial performance analysis

The executive and sales dashboards provide a clear view of business scale, revenue performance, order behavior, category contribution, payment methods, and market distribution.



Executive Overview

- Total Orders: **4,646**
- Total Sellers: **649**
- Total Customers: **4,622**
- Total Revenue: **19.77M**

Sales Performance

- Total Sales: **23.02M**
- AOV: **4.96K**
- Basket Size: **1.09**
- Daily Avg Orders: **7.73**
- Cancelled Rate: **0.60%**

Commercial Analysis

- Hourly order heatmap
- AOV trend by year and month
- Sales by payment method
- Pareto analysis by category

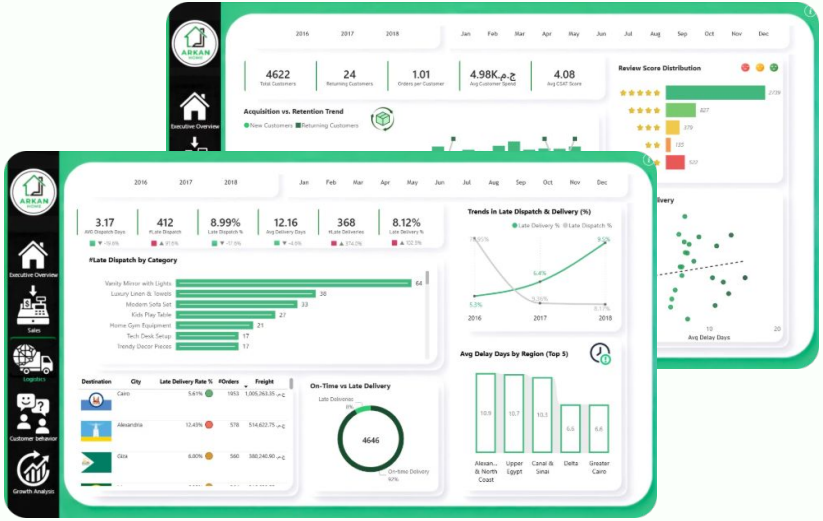
Key Message: Executive and sales analytics translate transactions into commercial intelligence. They help stakeholders understand performance, revenue drivers, and market activity.

OPERATIONAL INTELLIGENCE

Logistics & Customer Behavior Insights

Connecting operational performance with customer experience

The logistics and customer behavior dashboards help connect fulfillment performance, delivery reliability, acquisition patterns, retention signals, review scores, and customer satisfaction.



Logistics Performance

- Avg Dispatch Days: **3.17**
- Late Dispatch %: **8.99%**
- Avg Delivery Days: **12.16**
- Late Delivery %: **8.12%**
- Late Deliveries: **368**

Customer Behavior

- Total Customers: **4,622**
- Returning Customers: **24**
- Orders per Customer: **1.01**
- Avg Customer Spend: **4.98K**
- Avg CSAT Score: **4.08**

Decision Support Value

- Identifies logistics bottlenecks
- Highlights customer satisfaction signals
- Connects delivery performance with customer experience
- Supports service improvement actions

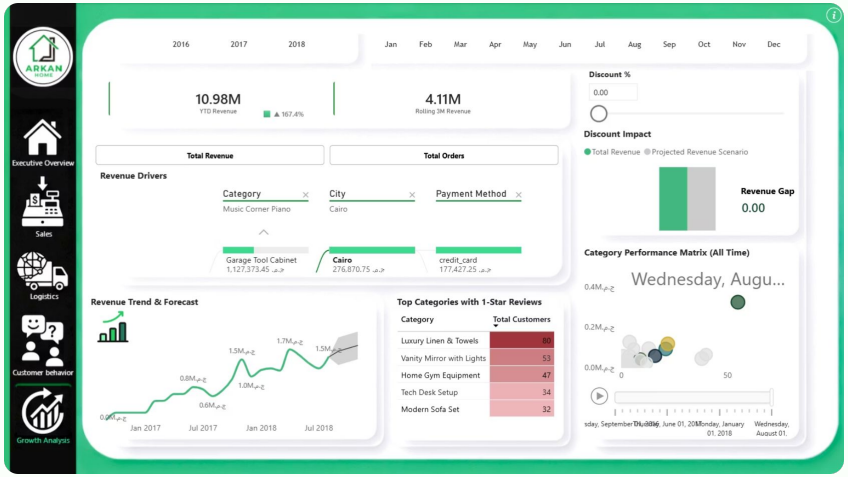
Key Message: Operational intelligence becomes stronger when logistics and customer behavior are analyzed together. This view connects delivery performance with satisfaction, retention, and service improvement opportunities.

STRATEGIC ANALYSIS

Growth Analysis & Decision Support

Turning business performance into strategic planning insights

The growth analysis dashboard supports strategic review by combining revenue trends, category contribution, discount scenario analysis, forecast movement, and performance risk indicators.



Growth Drivers

- Revenue driver decomposition
- Category, city, and payment method impact
- Top categories with weak review signals

Scenario Thinking

- Discount impact simulation
- Projected revenue comparison
- Revenue gap visibility
- Revenue trend and forecast review

Decision Support

- Supports growth planning
- Identifies revenue leakage areas
- Reveals category-level risks
- Connects performance with action priorities

Key Message: Growth analysis turns performance history into future-focused decision support. It helps compare revenue drivers, scenarios, and category risks in one analytical view.

TECHNICAL FOUNDATION

Technical Skills, DAX & Challenges

The technical foundation behind the final business intelligence solution

The project demonstrates a full BI workflow across data review, SQL engineering, analytical modeling, DAX logic, dashboard design, and business analysis.

Skills Applied

- **Excel & Data Review**
- **SQL Server**
- **Data Cleaning**
- **ETL**
- **Star Schema**
- **DAX**
- **Power BI**
- **Business Analysis**

DAX & KPI Logic

- Total Revenue
- AOV
- Basket Size
- Daily Avg Orders
- Late Delivery %
- Returning Customers
- Avg CSAT Score
- Projected Revenue Scenario

Challenges Solved

- Data quality inconsistencies
- Localization of a non-Egyptian dataset
- Modeling raw transactions into star schema
- Designing a usable multi-page dashboard experience

Key Message: The project proves capability across the full BI lifecycle. Technical execution and business analysis were combined into one complete decision-support solution.

GRADUATION PROJECT

Final Outcomes & Closing

From raw data to localized business intelligence

Arkan Home transformed raw e-commerce data into a complete localized BI solution that supports executive review, sales analysis, logistics monitoring, customer understanding, growth planning, and decision support.

End-to-End BI Solution
From raw data preparation to final dashboard storytelling.

Localized Business Case
Converted the original dataset into an Egyptian Home Decor and Furniture scenario.

Decision-Support Dashboard
Delivered interactive pages for executive, sales, logistics, customer, and growth analysis.

Team: Ahmed Magdy · Mark Morris · Mai Ibrahim · Aisha Taha

Thank You

From raw data to business intelligence. A complete analytics journey designed to support smarter decisions.